Operation Guide for the Use of Independent Serial Port Tools

Since some issues during the reproduction process of Xiaozhi AI require the use of serial port tools to locate problems, and it is impossible to analyze the specific causes of problems from photos or simple videos, in this case, we generally need to use serial port tools to display the information output by the UART interface of the development board. Here, we introduce the use of a relatively simple standalone serial port tool.

Developers can use the built-in serial terminal of VS Code or the IDF compilation tool to directly view serial log information, which will not be introduced here (experienced users may ignore this document).

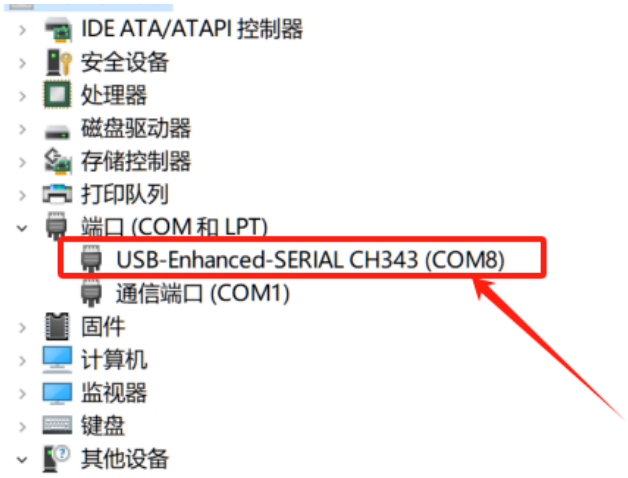
Method 1: Operating Instructions for Using Independent Serial Port Tool Software

1. Hardware and Software Preparation

1) Connect the computer to the UART interface on the right side of the ESP32-S3-WROOM-1 development board

Please first ensure that the ESP32-S3 development board is connected to your computer. In**My Computer - Device Manager**, you can see the COM port, as shown in the figure below. If you remember the Port Number, simply open the serial port tool and select the corresponding COM port.

Note: The COM8 port shown in the figure below is for reference only. Please select the port of your development board based on the COM port displayed on your computer, which may not necessarily be COM8. **The COM port recognized by your computer shall prevail.**



2) Serial Port Tool Download

If you have other serial port tools, you can use them directly. The operation steps are roughly the same, and there is no need to download them again.

If you don't have a serial port tool, you can download the Anxinke Serial Port Tool (no installation required, just double-click to open and use) from the official QQ group files of Xiaozhi AI:

Can be directly downloaded; downloading tool files may require a short wait:

2. Serial Port Tool Operation Steps

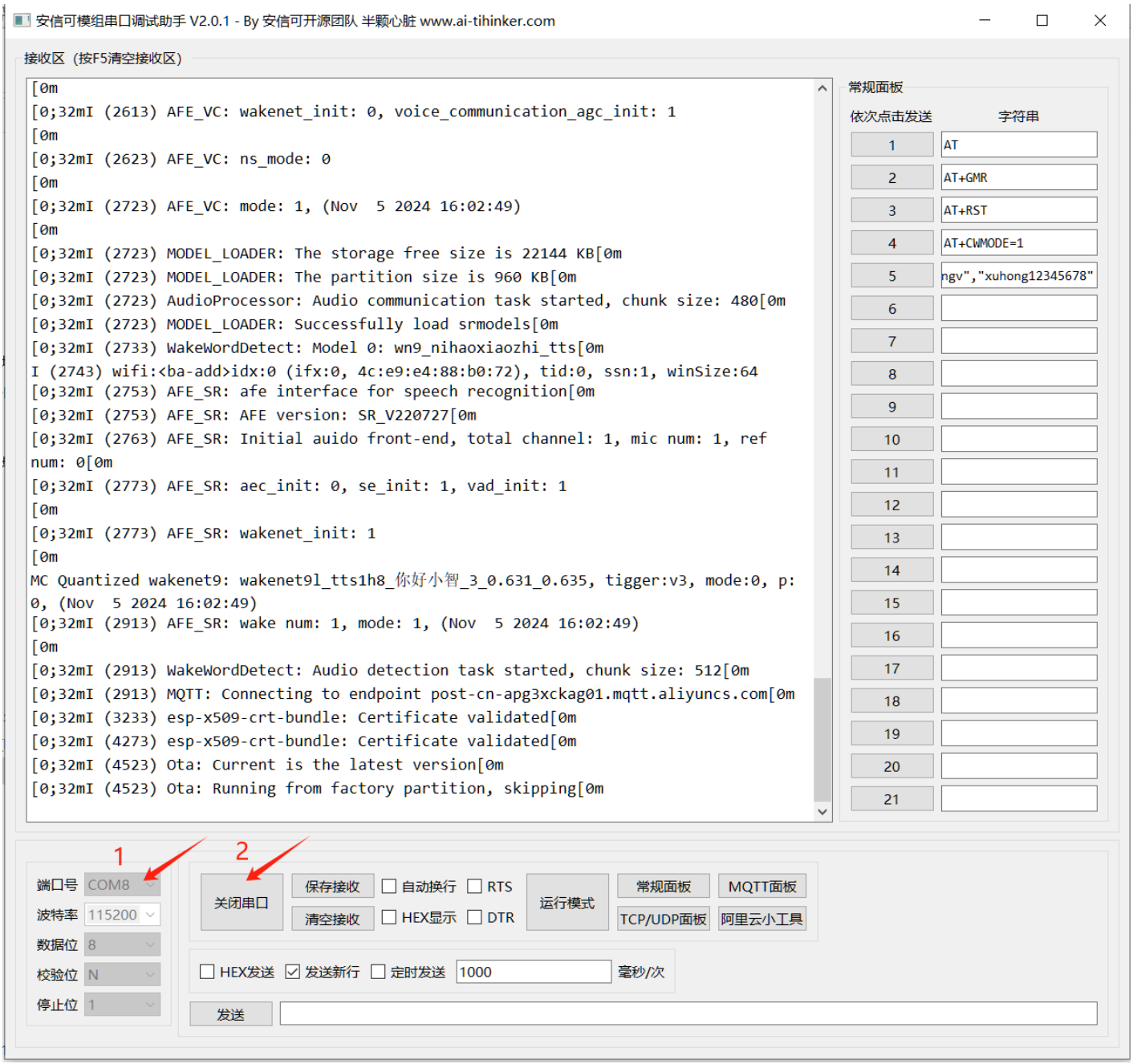
Step 1: Select the COM Port

Please select the corresponding COM port based on the COM Port Number of the development board/product such as ESP32-S3 or C3 identified in the first step.

Step 2: Click to open the serial port

Click the**Open Serial Port**button next to the COM Port Number, wait for the serial port information to be output. If necessary, you can press the RST button on the development board to restart the development board and view the serial port loading information from the beginning. Other configurations can remain default and do not need to be changed.

Note: When opening the serial port, please ensure that the target COM Port Number is not connected by other serial port tools; otherwise, a prompt indicating that the serial port is occupied and you do not have permission will be displayed.



Note: If you use other serial port tools or change the configuration of the serial port tool, and garbled characters or Chinese characters do not display, please uncheck the HEX display option. The default baud rate is 115200. You can explore and study the use of more serial port tools on your own.

Method 2: Third-party Online Serial Port Debugging Tool

1. Online Serial Port Terminal Debugging Tool - Ludaima IoT

For a better experience, please use the Edge browser to access. It comes with examples of common commands, and the debugging interface reference is as shown in the following illustration:

https://dev.16302.com/tools/terminal